P05

Random Walks on Electric Networks and Commute Times

Authors P. G. Doyle, J. L. Snell

Published Random Walks and Electric Networks (1984)

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Domain DOM_MATH

Abstract

We establish the equivalence between discrete-time random walks on reversible Markov chains and currents in resistive networks. Effective resistance measures the commute time between vertices.

Abstract quoted from P. G. Doyle et al., Random Walks and Electric Networks (1984).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?